

EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making.

Algorithm 1

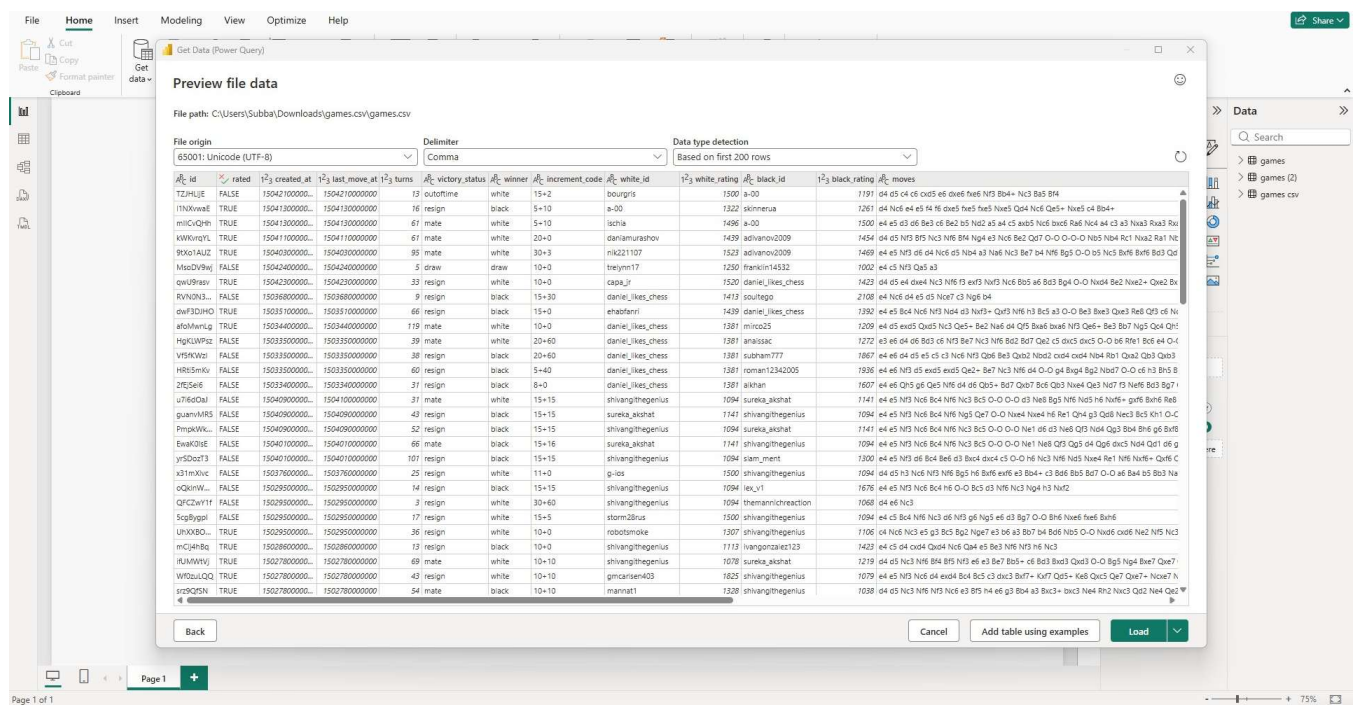
Importing Data into Power BI

Step 1 Open Power BI Desktop

Step 2 Select: Home → Get Data

Step 3 Choose CSV

Step 4 Browse dataset and select it



Algorithm 2

Data Transformation using Power Query

Step 1 Select Transform Data

Step 2 Open Power Query Editor

Step 3 Remove duplicates

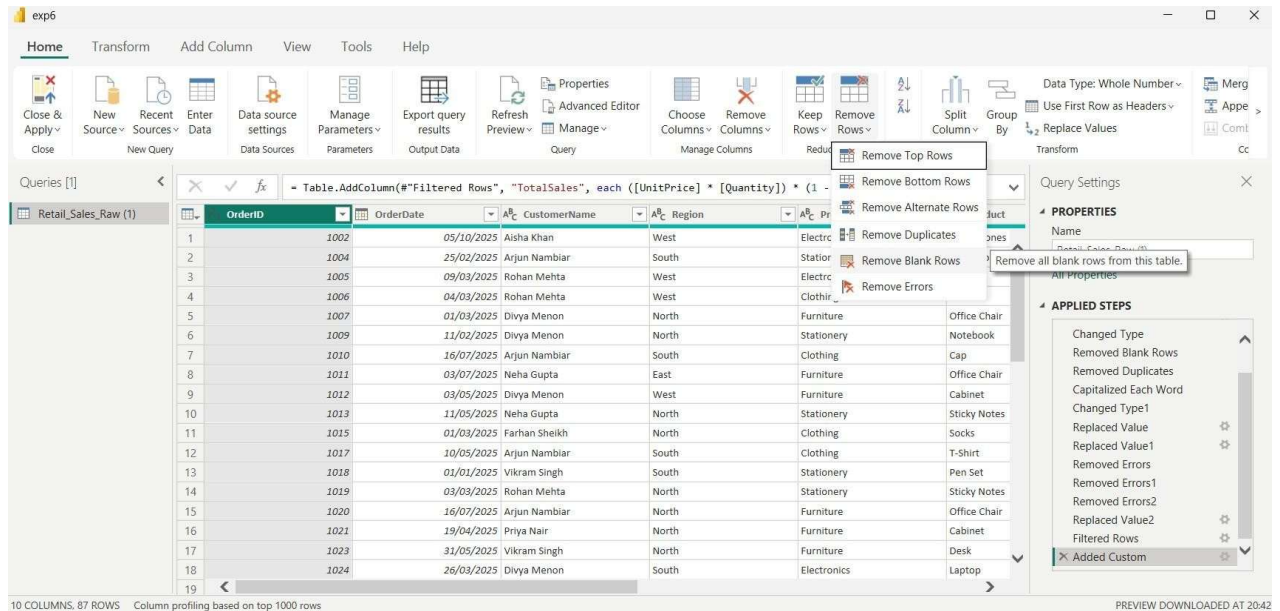
Step 4 Handle null values

Step 5 Rename columns

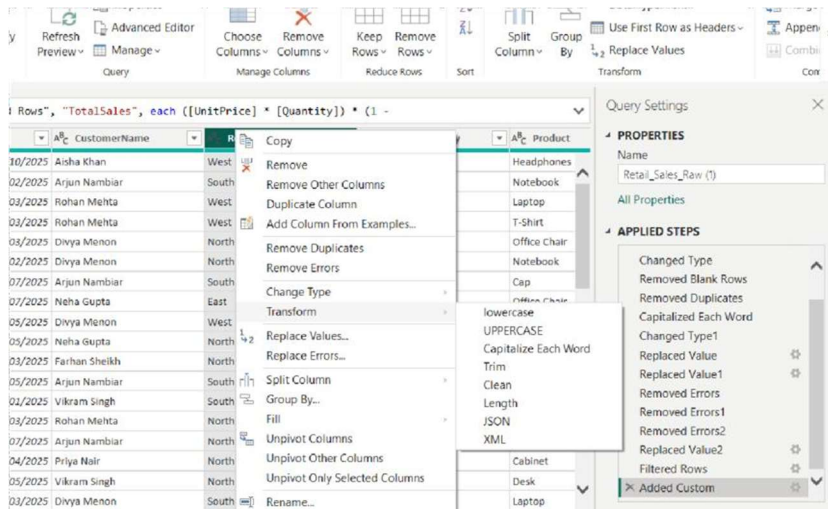
Step 6 Apply changes

Procedure

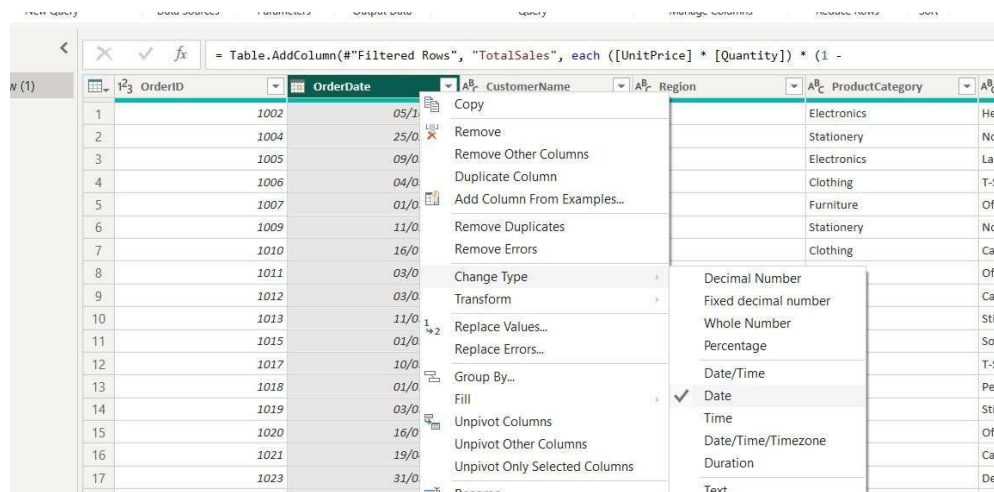
1. The data preview window will appear. Select Transform Data instead of Load to open the Power Query Editor.
2. In the Power Query Editor, examine the imported table and verify that all required columns are present, such as OrderID, OrderDate, CustomerName, Region, Quantity, UnitPrice, and Discount.
3. Remove blank rows from the dataset by selecting Home → Remove Rows → Remove Blank Rows. This removes completely empty records that could interfere with analysis and calculations.



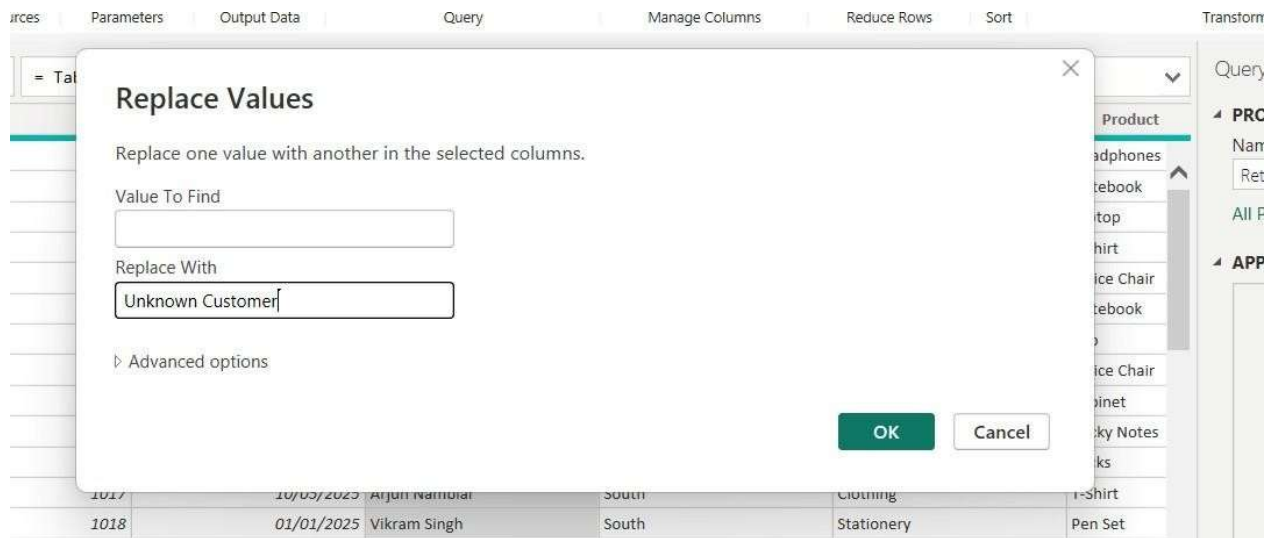
4. Remove duplicate records by selecting all columns in the table using Ctrl+A on the column headers and choosing Home → Remove Rows → Remove Duplicates. Power Query will remove records that contain identical values across all selected columns.
5. Standardize the values in the Region column. Right-click the Region column, select Transform, and choose Capitalize Each Word. This ensures that values such as “north”, “North”, and “NORTH” are represented consistently as “North”.



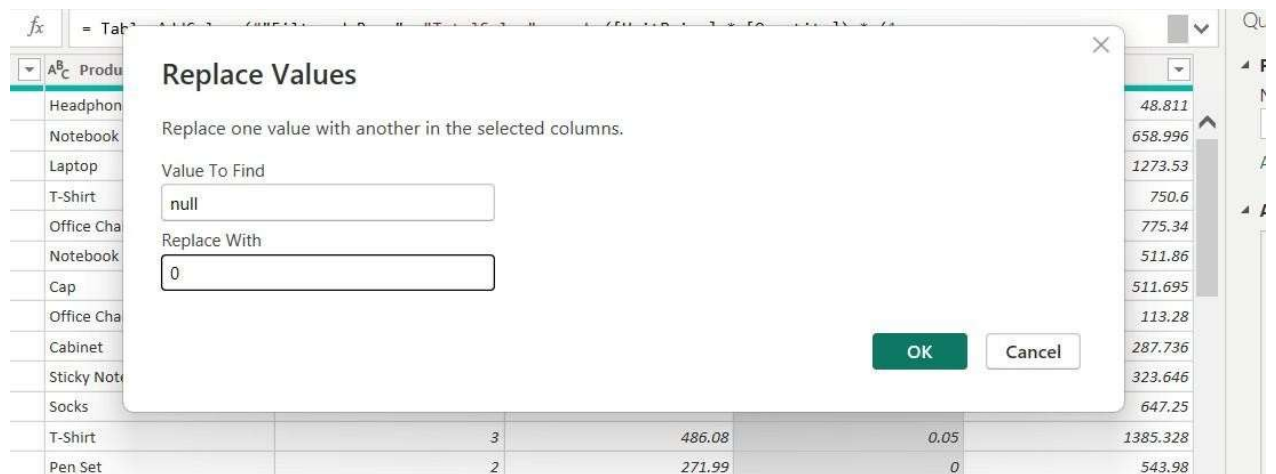
6. Standardize the OrderDate column by selecting it and choosing Transform → Data Type → Date. If Power Query displays errors because the dates use an inconsistent format, right-click the OrderDate column, select Change Type → Using Locale, and choose the appropriate date format and locale.



7. Handle missing values in the CustomerName column. Right-click CustomerName, select Replace Values, and replace blank or null values with "Unknown Customer".



8. Handle missing values in the Discount column. Replace null values with 0 using the Replace Values option. A value of 0 represents that no discount was applied when the discount information is missing.



9. Identify and remove invalid Quantity values. Select the Quantity column and apply a filter by pressing the arrow next to the column name to exclude records where Quantity is less than 0. Negative quantities are treated as invalid business data because a normal sales transaction cannot contain a negative quantity.

Quantity	UnitPrice	Discount
51.38	0.05	
173.42	0.05	
424.51	0	
150.12	0	
387.67	0	
269.4	0.05	
	0.1	
	0	
	0.05	
	0.05	
	0	
	0.05	
	0	
	0.1	

Filter Rows

Apply one or more filter conditions to the rows in this table.

☒ Basic ☐ Advanced

Keep rows where 'Quantity'

is greater than or equal to 0

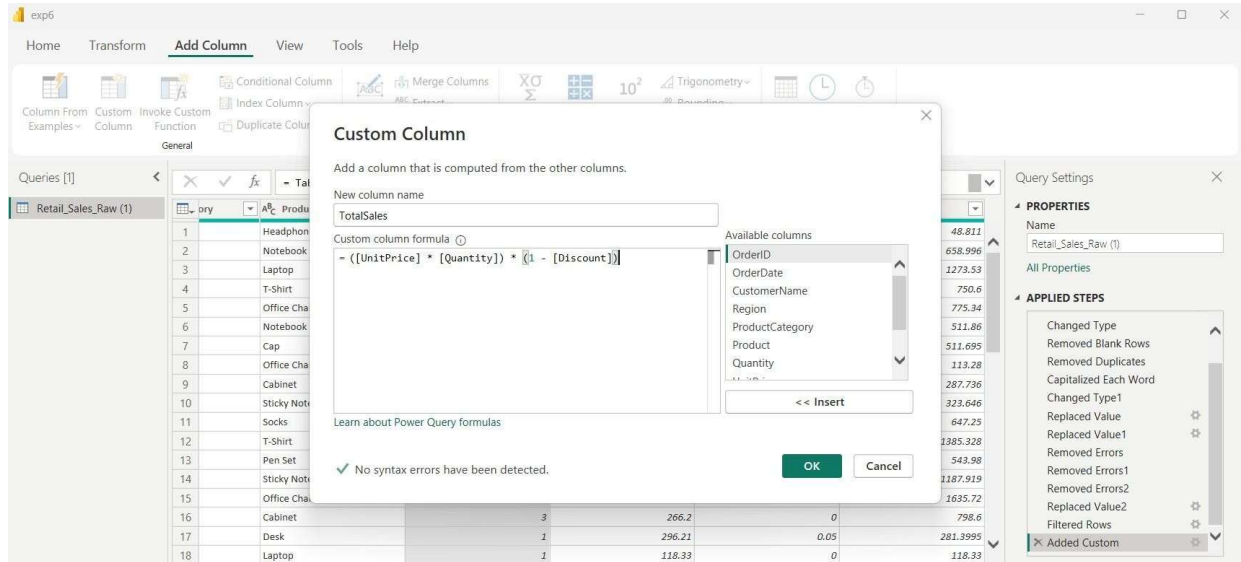
☒ And ☐ Or

Enter or select a valu...

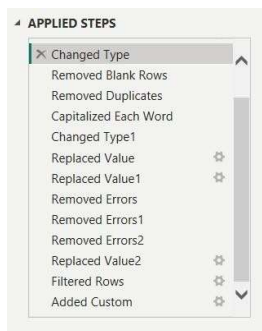
OK

Cancel

10. Create a derived column for total sales. Select Add Column → Custom Column and name the new column TotalSales. Enter the formula ([UnitPrice] * [Quantity]) * (1 - [Discount]). This calculates the actual sales value after applying the discount.



11. Verify and correct the data types of all columns. Set OrderID to Whole Number, OrderDate to Date, Quantity to Whole Number, UnitPrice to Decimal Number, Discount to Decimal Number, TotalSales to Decimal Number, and the remaining descriptive columns to Text.
12. Review the Applied Steps pane on the right side of the Power Query Editor. Confirm that all transformation steps have been recorded in the correct order. If an error is present, select the corresponding step and correct the transformation.



13. After completing all cleaning and transformation operations, select Home → Close & Apply. Power BI will apply the transformations and load the cleaned data into the data model.
14. Switch to the Data/Table view and inspect the transformed table. Verify that duplicate rows, blank rows, inconsistent Region values, invalid dates, missing customer names, missing discounts, and negative quantities have been handled correctly.

File origin	Delimiter	Data type detection
65001: Unicode (UTF-8)	Comma	Based on first 200 rows
Δ id	Δ rated	Δ created_at
Δ last_move_at	Δ turns	Δ victory_status
Δ winner	Δ increment_code	Δ white_id
Δ white_rating	Δ black_id	Δ black_rating
Δ moves		
TZHLJE	FALSE	1504210000000
1YNKwAE	TRUE	1504130000000
mICVQHh	TRUE	1504130000000
KWQvqYL	TRUE	1504110000000
9Xo1AUZ	TRUE	1504200000000
MsoDV9w	FALSE	1504240000000
qwU9rav	TRUE	1504230000000
RVN0N3	FALSE	1503680000000
owF3DjHD	TRUE	1503510000000
4b4kAwg	TRUE	1503440000000
HgIKU9Rz	FALSE	1503350000000
VfRWQ3	FALSE	1503350000000
HR15mV	FALSE	1503350000000
2fE5eIf	FALSE	1503340000000
u74dOxJ	FALSE	1504090000000
guarvMRS	FALSE	1504090000000
PmpKWL	FALSE	1504090000000
EwaK0sE	FALSE	1504010000000
jr5DazT3	FALSE	1504010000000
x31mXvC	FALSE	1503780000000
Qd0mW...	FALSE	1502950000000
QPCZwYF	FALSE	1502950000000
SqgBypt	FALSE	1502950000000
UXVXB0	TRUE	1502950000000
mC4HhB	TRUE	1502860000000
WU0uLQW	TRUE	1502780000000
Wf0uLQW	TRUE	1502780000000
sr2r5fSN	TRUE	1502780000000
outoftime	white	15+2
resign	black	5+10
mate	white	5+10
mate	white	20+0
mate	white	30+3
draw	draw	10+0
resign	white	10+0
black	black	15+30
black	black	15+0
black	black	10+0
black	black	20+60
black	black	20+60
black	black	5+40
black	black	8+0
mate	white	15+15
black	black	15+15
black	black	15+15
black	black	15+16
black	black	15+15
white	white	11+0
black	black	15+15
white	white	30+60
white	white	15+5
white	white	10+0
black	black	10+0
white	white	10+10
white	white	10+10
black	black	10+10

Algorithm 3

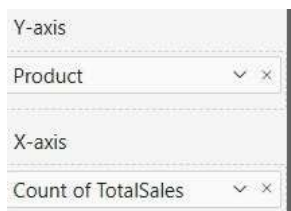
Creating a Bar Chart

Step 1 Choose Bar Chart Visualization



Step 2 Drag Product → Y-Axis

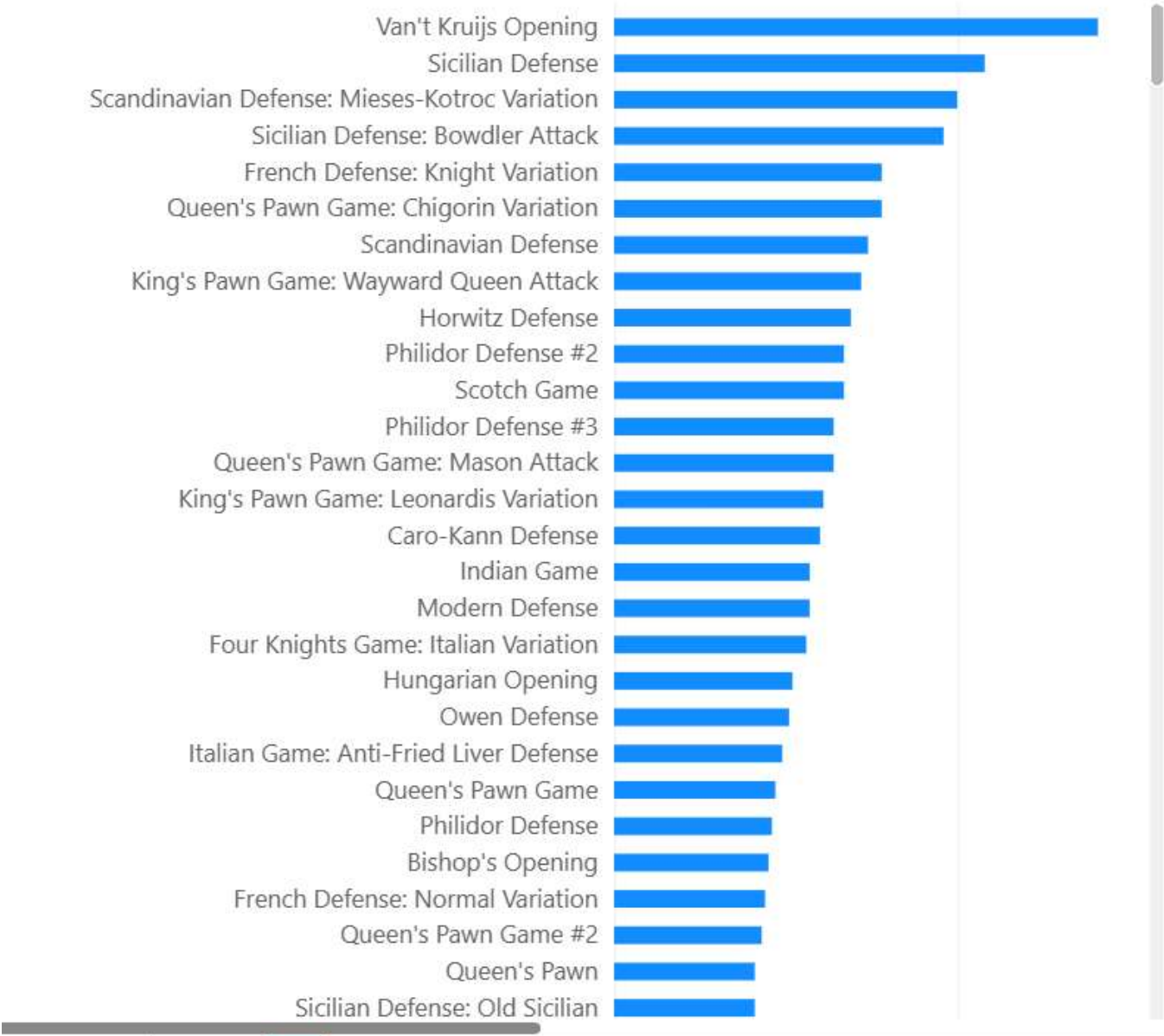
Step 3 Drag TotalSales → X axis



Step 4 Format chart

Count of TotalGames

by opening_name



Interpretation:

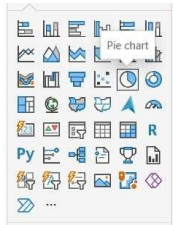
Cabinet generated highest sales

Stapler generated lowest sales

Algorithm 4

Creating a Pie Chart

Step 1 Select Pie Chart



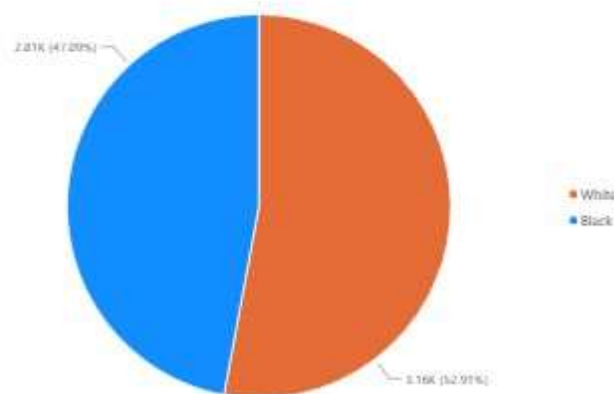
Step 2 Drag Product → Legend

Step 3 Drag TotalSales → Values



Output:

Count of TotalGames
by winner



Interpretation:

Cabinet generated highest Sales with 11.49%

Algorithm 5

Creating a KPI Card

Step 1 Create 4 measures by clicking on the New Measure from Measure tools tab, type the formula inside the dax field:

a. TotalSalesMeasure

The screenshot shows the 'Measure tools' ribbon in Power BI. The 'Name' field is set to 'TotalSalesMeasure', the 'Home table' is 'Retail_Sales_Raw (1)', the 'Format' is 'Whole number', and the 'Data category' is 'Uncategorized'. The DAX formula bar contains the formula: `1 TotalSalesMeasure = SUM('Retail_Sales_Raw (1)')[TotalSales]`.

b. Minimum

The screenshot shows the 'Measure tools' ribbon in Power BI. The 'Name' field is set to 'Minimum Sales', the 'Home table' is 'Retail_Sales_Raw (1)', the 'Format' is 'Whole number', and the 'Data category' is 'Uncategorized'. The DAX formula bar contains the formula: `1 Minimum Sales = MIN('Retail_Sales_Raw (1)')[TotalSales]`.

c.

Maximum

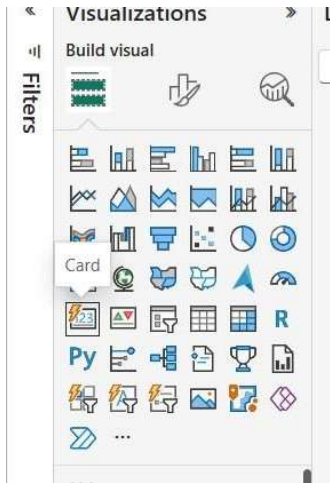
The screenshot shows the 'Measure tools' ribbon in Power BI. The 'Name' field is set to 'Maximum Sales', the 'Home table' is 'Retail_Sales_Raw (1)', the 'Format' is 'Whole number', and the 'Data category' is 'Uncategorized'. The DAX formula bar contains the formula: `1 Maximum Sales = MAX('Retail_Sales_Raw (1)')[TotalSales]`.

d.

Average

The screenshot shows the 'Measure tools' ribbon in Power BI. The 'Name' field is set to 'AverageSalesMeas...', the 'Home table' is 'Retail_Sales_Raw (1)', the 'Format' is 'General', and the 'Data category' is 'Uncategorized'. The DAX formula bar contains the formula: `1 AverageSalesMeasure = AVERAGE('Retail_Sales_Raw (1)')[TotalSales]`.

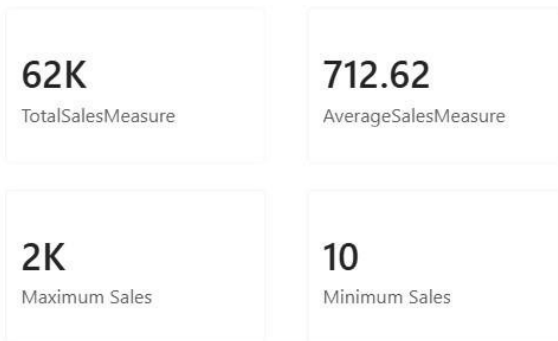
Step 2 Create card visual



Step 3 Drag that measure into the field



Output:



Algorithm 6

Creating Slicers

Step 1 Select Slicer Visualization

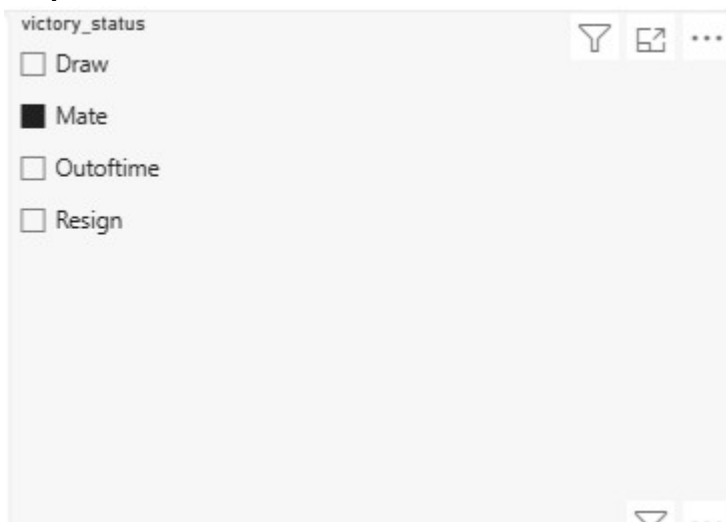
Step 2 Drag Region Field



Step 3 Enable Single Selection

Step 4 Apply Filters

Output:



Algorithm 8

Creating Dashboard

Step 1 Add KPI Card

Step 2 Add Bar Chart

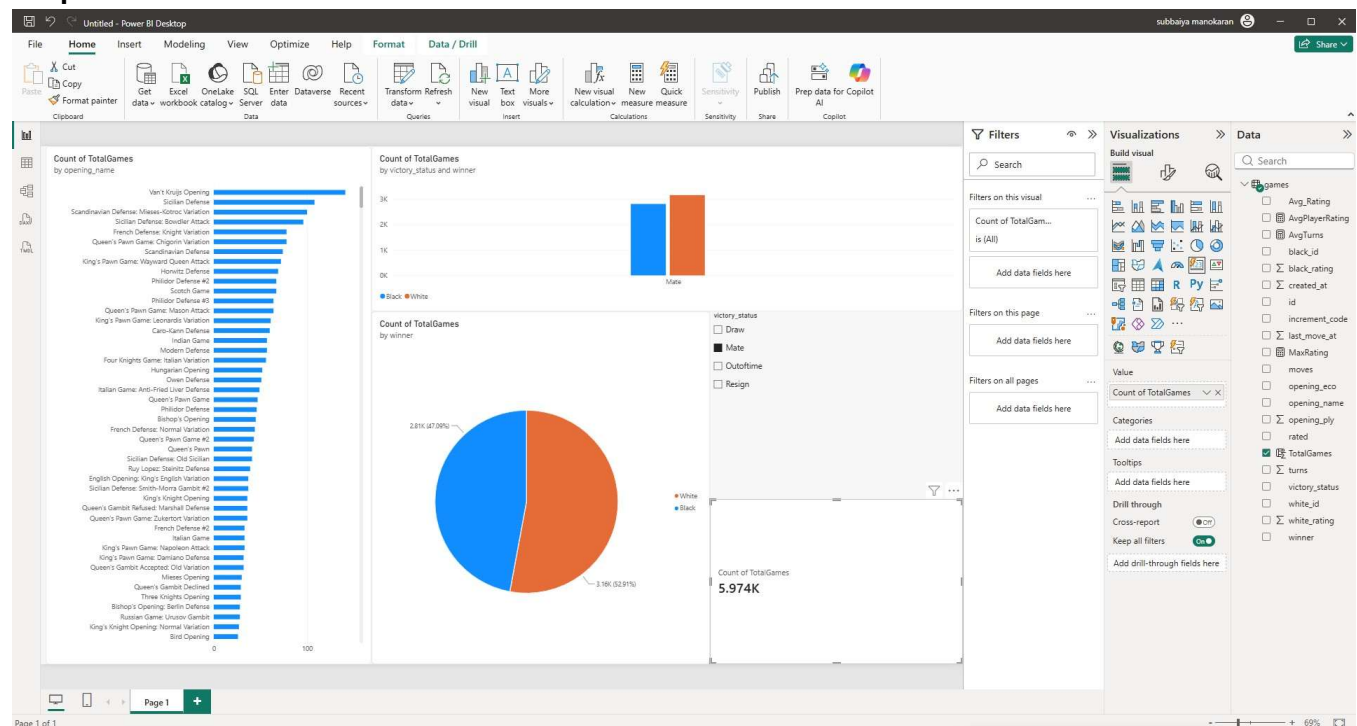
Step 3 Add Pie Chart

Step 4 Add Slicer

Step 5 Arrange Visuals

Step 6 Save Dashboard

Output:



Result:

The data was successfully imported, transformed, analyzed, and visualized in Power BI Desktop, and an interactive dashboard was created for effective decision-making.